

Quantum Generative Augmentation for Novel-Pattern Fraud Response

the first five confirmed cases — where a bank is blind and the losses compound

Answering the challenge statement section by section · every number carries a cloud job receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-hsbc-gic2026>

Executive summary

+5.8 pts

more fraud caught at the **same alert budget** when only **five confirmed cases** of a new pattern exist (66.4% vs 60.6% for SMOTE) — 40 seeds, paired **p = 0.001**, and the outcome **variance halves** (± 4.3 vs ± 8.2 frauds)

\$5–13M

per year at ~\$300bn annual card volume, built from the measured uplift and the challenge's own anchors (\$4.41 total cost per \$1 of fraud). Time-to-response value — the sensitivity model ships so HSBC can substitute its own parameters

$n \leq 5$

the window, characterised rather than claimed: decisive at five labels, gone by ten, reversed beyond twenty — exactly what the *power of data* result predicts. We report the regimes where classical wins, with the same receipts

Why the numbers can be trusted

One referee — full network physics (DC-PTDF thermal congestion plus LinDistFlow voltage) — grades every solver, classical and quantum, so all comparisons share one standard. Every claim carries a frozen pre-registration written before execution and a cloud job ID across IBM Heron and QCi Dirac-3. Hard-instance difficulty is established as a **measured convergence law**, not a single timeout: on the 26-variable instance the proof gap closes to 22.6% after 2.6 hours and extrapolates to 17–21% still unproven after a full afternoon, with exhaustive certification costing 87 CPU-days. A credential-free audit (`python verify.py`) replays every headline number from archived data in about three minutes.

Scalability is demonstrated rather than projected: the same operator family runs from a 20-qubit NISQ instance up to **64 rungs — 128+ qubits — on IBM Heron**, bracketing the brief's requested range with one algorithm. All circuits are standard Qiskit; the E.ON team can re-execute and validate every hardware result directly.

Technical report follows (§0–§8) · Reproducibility & Receipts companion document supplied separately · <https://github.com/sharadbachani-oss/merlin-quantum-hsbc-gic2026>

Technical Report

The result in one line. When a new fraud pattern is first detected and only **five confirmed cases** exist, augmenting the training set with samples drawn from a quantum device catches **+5.8 percentage points more fraud** than SMOTE at the same alert budget (66.4% vs 60.6%, 40 seeds, $p = 0.001$) — and halves the variance of that outcome. The advantage is confined to the few-label regime and disappears by ten cases, which is exactly where theory says a quantum advantage on classical data can exist.

Response to §4.1 — Primary objective

The problem we target. A bank's steady-state fraud model is not where the pain is: on mature patterns with thousands of labels, gradient boosting is near its ceiling and no method meaningfully improves on it — we measured this and report it in §4.2. The operational pain is the **novel-pattern window**: a new typology surfaces, the bank holds a handful of confirmed cases, and every day before an effective model deploys is uncontrolled loss.

The component. A hybrid pipeline in which the quantum device performs **generative synthesis of minority-class training samples** — the pipeline role the statement explicitly invites ("use quantum computation for a specific component of the pipeline... feature transformation, kernel estimation, anomaly scoring, or re-ranking"). Inference remains entirely classical.

Datasets. ULB European Cardholder (284,807 transactions, 492 frauds, 0.172%) as the primary — the statement's own extreme-imbalance benchmark, where its authors specify AUPRC (Dal Pozzolo et al., 2015). IEEE-CIS (590,540) and Sparkov (1.3M) for the saturation and scale controls in §4.2.

Classical baselines (tuned XGBoost, temporal 80/20 split — the harder and operationally honest evaluation, addressing the statement's distribution-shift consideration):

dataset	AUC-ROC	AUPRC	F1
ULB (284k, temporal)	0.9838	0.8065	0.8321
IEEE-CIS (590k, temporal, identity-merged)	0.9118	0.5532	0.5462

Response to §4.1 — The measured result

Setup. Five confirmed fraud cases drawn at random from the training period; the classifier trained on those five plus 2,000 synthetic minority samples from each augmentation method; evaluated on the untouched temporal test split (56,962 transactions, 75 frauds). Forty independent draws of the five cases; paired statistics throughout.

augmentation	AUPRC	worst case (5th pct)	frauds caught @0.2% alert budget	detection rate
none	0.5136 ± 0.120	0.2919	42.6 ± 8.4	56.8%
jitter	0.5146 ± 0.142	0.2904	42.9 ± 9.4	57.2%
SMOTE (industry standard)	0.5500 ± 0.122	0.3345	45.4 ± 8.2	60.6%
quantum generative	0.6068 ± 0.080	0.4676	49.8 ± 4.3	66.4%
comparison	AUPRC	worst case	frauds caught	p (paired)
vs SMOTE	+10.3%	+39.8%	+4.4	0.0011
vs jitter	+17.9%	+61.0%	+6.9	<0.0001
vs no augmentation	+18.1%	+60.2%	+7.2	<0.0001

Three properties make this operationally meaningful rather than a metric artefact:

1. **It translates to caught fraud.** +4.4 of 75 frauds at a *fixed* alert budget ($p = 0.001$) — no additional analyst load. 2. **It halves the variance** (± 4.3 vs ± 8.2 – 9.4). On a novel pattern the bank gets one attempt, not an average. SMOTE's 5th-percentile outcome is AUPRC 0.33; the quantum arm's is 0.47. 3. **It beats every classical arm**, including jitter — which outperforms the quantum arm in the data-rich regime (§4.2). The ordering inverts precisely where theory says it should.

Response to §4.2 — The conditions under which quantum helps

This is the statement's explicit request ("characterize under what conditions quantum approaches perform differently"), and we answer it with a measured curve rather than a claim.

labelled frauds	quantum vs SMOTE	significance
5	+10.3%	p = 0.001
10	+1.8%	p = 0.42 (not significant)
20	−3.5%	classical ahead
50	−3.3%	classical ahead
100	−1.3%	classical ahead
417 (full)	+0.6%	parity

The window is sharp: decisive at five labels, gone by ten, reversed beyond twenty. This is the behaviour the *power of data* result predicts (Huang et al., Nature Communications 2021): with classical data, classical models close any quantum gap as labels accumulate. The corollary — that the quantum window is where labels are scarce — is what we measure here.

Mechanism. With five examples, SMOTE can only interpolate inside the convex hull of those five points; Gaussian jitter perturbs them independently. Neither can place mass on regions the five samples do not span. The device samples a Gibbs distribution over an energy landscape fitted from the data's measured moments — including **1,306 three-body interaction terms** that a classical pairwise Boltzmann machine cannot represent, because higher-order partition functions are intractable to normalise. The sampler's intrinsic temperature supplies exploration that interpolation cannot.

Response to §4.2 — Where the approach does not help

Reported with the same receipts as the positive result.

- **Data-rich regime.** At 417 labels, Gaussian jitter (0.8177) beats the quantum arm (0.8084). We make no claim of steady-state superiority.
- **Saturated baselines.** On Sparkov the classifier reaches AUPRC 0.9238 / AUC 0.9985 unaugmented; SMOTE itself *degrades* it (0.9191) and the quantum arm is flat (0.9247 vs 0.9254). Augmentation is a data-starved-regime tool; we measured that boundary rather than assuming it.
- **Quantum feature maps.** A 128-qubit feature-extraction card applied to the classifier's ambiguous decision band produced +0.0023 AUPRC — within seed noise (random-feature control: −0.0284). Discrimination is not where the advantage lives.
- **Data-encoded Boltzmann machine on gate hardware.** Fields and couplings read directly from fraud moments onto 87 qubits (1,152 two-qubit gates, depth 196) gave 3-body moment error 0.2616 against 0.0033 for the best classical generator — and its own fields-zeroed control scored *better* (0.0587). Decoherence dominates before the encoded landscape shapes the distribution. This architecture needs deeper coherence than current hardware provides.

Response to §4.2 — Hardware execution

platform	role	scale	receipt
QCi Dirac-3	degree-3 generative sampling (scored result)	24 variables, 600-term landscape, 354 jobs, 4,803 samples	results/ulb_raw400.json
IBM Heron ibm_fez	128-qubit generative sampling	65 circuits × 8,192 shots, 2,656 two-qubit gates, 524,288 samples in one job	dab6therrl7c7386flo0
IBM Heron ibm_fez	128-qubit feature extraction	206 circuits × 8,192 shots, 2,720 two-qubit gates	daaucejvpcac73dd232g
IBM Heron ibm_fez	data-encoded Boltzmann machine	87 qubits, 1,152 two-qubit gates, depth 196	daba21jvpcac73ddh9h0
QuEra Aquila (Braket)	256-atom analog anomaly encoding	8 jobs, 99.3% mean atom loading	Braket console

Sampling-diversity ceiling, measured. The Dirac arm was run at 40 and at 354 jobs: 82 versus 124 distinct patterns — **8.8× the device time for 1.51× the diversity**. Sampling breadth is set by the device's intrinsic relaxation temperature, not by job count, so the reported result is a floor achieved at 82 patterns rather than a favourable draw from a long run. The same workload on IBM Heron produced 524,288 samples from a single job.

Noise handling and simulator-versus-hardware. Dynamical decoupling (XpXm) on every gate-model circuit; one parameterised template transpiled once with a fixed layout and seed, per-sample circuits produced by angle re-binding so every sample rides the identical physical circuit; interleaved null references and in-job replica pairs (**replica agreement 0.0039**). Encoding mechanics were rehearsed end-to-end on Braket's local AHS simulator before hardware submission (variability 0.076, data-versus-null separation 0.111, shot-split reproducibility 0.045).

Scale relative to the published bar. Published quantum generative work for fraud (Q-SYNTH arXiv:2605.21164; QCBM tabular augmentation arXiv:2607.09113) uses *variational* generators — trained circuits that encounter barren plateaus and operate at ~10–20 qubits. Both quantum arms here are **training-free**: the Dirac landscape is read directly from data moments; the Heron generator has no variational parameters. The 128-qubit run produced half a million samples from one job, roughly a 6× qubit-scale increase over the published state of the art for this application. Separately, the Aquila execution held 99.3% mean atom loading across 8 jobs where the closest comparable published work on the same processor (Vitali et al., arXiv:2510.04747) reported noise-dominated failure.

Response to §5.2 — Outputs

- **Fraud probability** (float [0,1]) and **binary prediction** per transaction from the augmented classifier; confusion matrices, precision and recall for every arm in `results/`.
- **Feature attribution.** SHAP values per prediction classically; and on the quantum side the generative landscape is *directly interpretable* — every term is a named literal or an interaction among named literals carrying a measured log-odds coefficient, so any synthetic sample traces to the specific fraud statistics that produced it. That is a model-governance property rather than a post-hoc explanation.
- **Comparison with classical baselines.** Tuned XGBoost plus four augmentation arms, full metric slate, 40 seeds, paired tests.
- **Encoding, circuit design, conditions.** §4.2 and §5.3.

Commercial value

Applied to the novel-pattern window, with every input either measured or sourced:

input	value	source
detection uplift	+5.8 pp (60.6% → 66.4%)	measured, 40 seeds, $p = 0.001$
card fraud rate	6.8 bps of volume	Nilson Report 2025
total cost per \$1 fraud	\$4.41	LexisNexis 2024 (the statement's anchor)
share of fraud in the few-label window	10–25%	assumption — HSBC's own figure should replace it

\$5M–\$13M per year at ~\$300bn annual card volume (\$1.7M–\$4.3M per \$100bn; scales linearly). Stated precisely: this is **time-to-response value, not steady-state detection improvement** — it accrues only while a pattern has ≤ 5 confirmed cases. The variance halving may be worth more than the mean: on a single emerging attack the bank gets one draw, and avoiding one bad draw can exceed the annualised figure.

The sensitivity model ships in the package so HSBC can substitute its own volume and novel-pattern share.

Response to §5.3 — Deployment considerations

Latency. Quantum computation is training-time only; inference is the classical model at sub-millisecond per transaction, against the 100–300 ms authorisation budget the statement describes. No quantum hardware sits in the decision path — the correct posture for a system that must degrade gracefully.

Class imbalance. Scale-positive weighting on all arms; 2,000 synthetic minority samples per augmentation arm; temporal splits preserve the natural fraud rate; no test-set resampling.

Feature selection for encoding. 24 binary literals from quantile thresholds on the most fraud-separating features (Dirac); 16 continuous features mapped to preparation angles and local tilts (Heron). No attempt is made to encode hundreds of features, per the statement's guidance.

Circuit design. Native heavy-hex geometry with zero SWAP overhead; edge-coloured layers for gate parallelism; fixed layouts selected against a measured device atlas rather than reported calibration.

Reproducibility

Every result carries a pre-registration written before execution, ≥ 3 seeds (40 for the headline), in-job controls, and cloud job IDs. Every headline number replays from archived data with `pip install numpy scikit-learn xgboost && python verify.py` — no credentials required. Negative results ship at the same receipt class as positive ones.

Team Merlin Digital — the measured result, the window it lives in, and the receipts for both.